

AHMED FERGANEY

Process Engineer | Operational Excellence & Manufacturing

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PROFILE

Process Engineer with hands-on experience in defect analysis, root cause analysis, and process improvement across automotive, electronics, and home appliance manufacturing lines. Proven record of raising First Pass Yield (FPY) and line efficiency, developing and updating SOPs and work instructions, and supporting new product introduction (NPI) and mass-production troubleshooting. Skilled in Lean Six Sigma, PFMEA/CAPA, and structured problem-solving (DMAIC, 5-Why, Fishbone), with experience collaborating across Production, Quality, and Engineering teams.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department
- Graduation Project: Design, Modeling, and Control of a Six-DOF Artificial Upper Limb

PROFESSIONAL EXPERIENCE

Product Development Engineer | Kandil Glass | 10th of Ramadan City, Egypt06/2026 – Present

- Led end-to-end **NPI for customized glass products**, from client requirements and feasibility analysis through design and full-scale production.
- Developed a **Power BI manufacturing analytics dashboard** integrating production, quality, process-loss, and supervisor-log data from multiple Excel sources.
- Designed a **galaxy-schema data model** covering production lines, products, customers, shifts, defects, job changes, and downtime reasons.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt02/2025 – 05/2026

- Led **Operational Excellence projects** across Assembly, Sheet Metal, Injection Molding, and Warehousing, implementing **SMED, 5S, Standard Work, Visual Management, and Daily Management Meetings** to improve productivity and process flow.
- Led a **Standard Work and line balancing project**, improving **LOB from 60% to 75%** and enhancing workload distribution.
- Supervised and coached production/process teams through **training, performance reviews, and cross-functional improvement initiatives** to build capability and sustain results.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt12/2023 – 01/2024

- Reduced **defect rate from 8% to 2% (75% reduction)** on a VW automotive electronics assembly line using **RCA, PFMEA, and CAPA**, improving FPY, throughput, and downtime.
- Applied **DMAIC, DFMEA, 5 Why, and Fishbone** to eliminate recurring failures, update SOPs/work instructions, and verify corrective-action effectiveness.
- Coordinated with **internal and external suppliers** on defect containment and prevention plans to reduce recurrence and protect line performance.

Continuous Improvement Engineer | JAC | Badr City, Egypt03/2023 – 11/2023

- Increased **Water Dispenser line productivity from 150 to 225 units per 2-hour cycle (50%)** through process and workflow optimization.
- Drove **continuous improvement** across Water Dispenser and TV production lines, identifying and closing efficiency gaps.

CORE QUALIFICATIONS

Continuous Improvement & Lean: Lean Manufacturing (Kaizen, 5S, SMED, Waste Reduction), Six Sigma fundamentals (DMAIC), Work Standardization & SOP Development, Line Balancing, Capacity Planning & Takt Time Analysis, Time & Motion Studies, Cycle Time Reduction

KPI & Performance Management: OEE, FPY, Scrap Rate, Downtime Reduction, Line of Balance (LOB) Improvement, Cost Reduction & Productivity Initiatives

Process Engineering: Process Design, Mapping & Optimization (VSM, SIPOC, Process Flow Diagrams), PFMEA/DFMEA & Control Plans, Root Cause Analysis (5 Whys, Fishbone/Ishikawa), New Product Introduction (NPI) Support, Production Planning & MRP Fundamentals (CPIM)

Quality Foundations: 8D, CAPA, RCA, Pareto Analysis, Audit Participation & Compliance Processes

Data & Analysis: Python for data analysis, SQL for production data querying, Statistical Process Control (SPC, Cp/Cpk, control charts, hypothesis testing), Power BI dashboards & data visualization

Leadership: Cross-functional Team Leadership, Team Supervision & Coaching, Training & Performance Reviews, Agile/Scrum Methodologies, Technical Mentoring, Problem-Solving Under Constraints

CERTIFICATIONS

- Oxyacetylene and ARC-Welding in High-Pressure Pipes Diploma | Don Bosco Institute | 2019
- Turning Course | Don Bosco Institute | 2020
- Automation Basic Diploma | HA Consulting Group | 2020
- Robotics Camp with a Case Study (Robotic Arm) | RTE | 2020
- Mechanics of Automotive Diploma | Don Bosco Institute | 2021
- Lean Six Sigma Yellow and Green Belt | Udemy | 2022
- CPIM Part One | HPA | 2023
- Embedded System Diploma | Eng. Ahmed Abd El-Ghafar | 2023
- Statistics, Linear Algebra & Numerical Optimization | ITI | 2024
- Databases, Power BI, Python for Data Analysis | Udemy | 2025
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, AI Agents | Udemy | 2026

PROJECTS

AdventureWorks Data Warehouse & Power BI Analytics | [GitHub](#)

- Designed a Kimball star-schema data warehouse (FactSales + 4 dimensions) from the AdventureWorks OLTP database.
- Built a full SSIS ETL package (Control Flow + Data Flow) to extract, transform, and load data; deployed to the SSIS Catalog and scheduled daily via SQL Server Agent.
- Developed Power BI dashboards with DAX measures (Revenue, YTD, YoY growth) covering sales trends, product performance, and territory analysis.

Customer Segmentation using Machine Learning | [GitHub](#)

- Segmented an automotive dataset using KMeans, Random Forest, and Linear Regression; applied feature engineering, outlier detection (DBSCAN), and PCA for actionable customer insights.

LANGUAGES & TOOLS

Languages: Arabic (Native), English (Professional), German (Basic)

Tools: AutoCAD, Autodesk Inventor, Power BI, MS Office, Jira, Trello, MATLAB